BOUNDED SUMS VERSUS BOUNDED DIFFERENCES: A NOTE ON ERDŐS PROBLEM #863

CLAUDE OPUS 5.5

Abstract. For $r \geq 1$ let $S_r(N)$ be the largest size of a set $A \subseteq \{1, \dots, N\}$ in which every integer has at most r representations $a + a'$ with $a, a' \in A$, $a \leq a'$, and let $D_r(N)$ be the largest size of a set $B \subseteq \{1, \dots, N\}$ in which every positive integer has at most r representations $b - b'$ with $b, b' \in B$. Erdős, recalling conversations with D. Berend and R. Freud, expected that for $r \geq 2$ the constants in $S_r(N) \sim c_r\sqrt{N}$ and $D_r(N) \sim c'_r\sqrt{N}$ satisfy $c'_r < c_r$. We observe that this follows from results in the literature: the Erdős–Turán argument for Sidon sets gives $D_r(N) \leq (\sqrt{r} + o(1))\sqrt{N}$ (indeed $D_r(N) \sim \sqrt{rN}$ was already known), while a construction of Cilleruelo, Ruzsa and Trujillo gives $S_r(N) \geq \big((r+s)/\sqrt{r+2s} - o(1)\big)\sqrt{N}$ with $s = \lfloor r/2 \rfloor$, and $(r+s)/\sqrt{r+2s} > \sqrt{r}$. We give a self-contained account with complete proofs of both bounds, and collect what is known about the two constants.

About this version. This paper was written by Claude Opus 5.5 (Anthropic) in October 2026. It is a complete rewrite of an earlier note posted by Boon Suan Ho in April 2026, whose results were obtained with GPT-5.4 Pro. **The original version of this paper is available from** `https://boonsuan.github.io/erdos863_old.pdf`**.** The mathematics is that of the original note, except that the upper bound is now proved with an explicit error term and the lifting lemma is stated for arbitrary layer patterns. This version adds figures, a survey of the known bounds, and corrected attributions; in particular, the asymptotic formula $D_r(N) \sim \sqrt{rN}$ was already in the literature.

1. Introduction

1.1. **The problem.** In his 1992 paper *Some of my forgotten problems in number theory*, Erdős recalls a question that he and D. Berend had noticed during a visit to Ben-Gurion University in Beer Sheva, and that he later reformulated independently with R. Freud before remembering the earlier conversation [Er92, pp. 39–40]. In the notation of the problem collection [Blo] it reads as follows.

Problem (Erdős Problem #863)**.** *Let $r \geq 2$. Let $A \subseteq \{1, \dots, N\}$ be a set of maximal size such that for every n there are at most r solutions of $n = a + b$ with $a, b \in A$, $a \leq b$, and let $B \subseteq \{1, \dots, N\}$ be a set of maximal size such that for every $n \geq 1$ there are at most r solutions of $n = a - b$ with $a, b \in B$. If $|A| \sim c_r N^{1/2}$ and $|B| \sim c'_r N^{1/2}$, is it true that $c_r \neq c'_r$? Is it true that $c'_r < c_r$?*

Sets of the first kind are the $B_2[r]$ *sets* (or generalized Sidon sets); sets of the second kind have been called *r-Golomb rulers* or $B_2^-[r]$ sets [CMT15]. For $r = 1$ the two conditions coincide: both say that A is a *Sidon set*, that is, all sums $a + a'$ with $a \leq a'$ are distinct, or equivalently all positive differences are distinct. So $c_1 = c'_1$, and the theorem of Erdős and Turán [ET41] (an upper bound $\sqrt{N} + O(N^{1/4})$) together with Singer's difference sets [Sin38] (whose use for this purpose Erdős credits to Chowla and himself [Er92, p. 37]) gives $c_1 = c'_1 = 1$. For $r \geq 2$ the

Date: October 2026.

conditions are different, and Erdős expected sums to allow larger sets. As evidence he quoted two results from his Hungarian paper with Freud [EF91]: if a single integer is allowed more than one representation as a sum, one can take about $\frac{2}{\sqrt{3}}\sqrt{N}$ elements, whereas the corresponding relaxation for differences still forces $(1+o(1))\sqrt{N}$. In Erdős's words these results "seem to give hope" for $c'_r < c_r$ [Er92, p. 40].

1.2. **The answer.** Write $S_r(N)$ and $D_r(N)$ for the two maxima (precise definitions are in Section 2). We prove:

Theorem 1.1. *Let $r \geq 2$ and $s = \lfloor r/2 \rfloor$. Then*

$$\limsup_{N\to\infty} \frac{D_r(N)}{\sqrt{N}} \;\leq\; \sqrt{r} \;<\; \frac{r+s}{\sqrt{r+2s}} \;\leq\; \liminf_{N\to\infty} \frac{S_r(N)}{\sqrt{N}}.$$

Corollary 1.2. *Let $r \geq 2$. If $S_r(N) \sim c_r\sqrt{N}$ and $D_r(N) \sim c'_r\sqrt{N}$, then $c'_r \leq \sqrt{r} < c_r$.*

So the answer to both of Erdős's questions is yes. More is true. It is known that $D_r(N) \sim \sqrt{rN}$ [CMT15, ST25] (see Section 6), so c'_r exists and equals $\sqrt{r}$, and Theorem 1.1 says that every limit point of $S_r(N)/\sqrt{N}$ is strictly larger than c'_r. Whether c_r itself exists is open for every $r \geq 2$ [CRV10, Pla01]. For $r = 2$, for instance, $c'_2 = \sqrt{2} \approx 1.414$, while the best lower bound for $S_2(N)/\sqrt{N}$ that we know of is $4/\sqrt{7} \approx 1.512$ [HP02] and the best upper bound we know of is 2.2851 [HP18]; see Table 1.

1.3. **What is new here and what is not.** Both halves of Theorem 1.1 are in the literature, and the purpose of this note is to point out that together they answer Problem #863, and to give a self-contained account.

- *The upper bound for differences* is the Erdős–Turán argument [ET41] for Sidon sets, which goes through unchanged for general r. In fact $D_r(N) \leq \sqrt{rN} + (rN)^{1/4} + 1$ is proved by Caicedo, Martos and Trujillo [CMT15, Thm. 2.5], who attribute it to earlier work of Trujillo, García and Velásquez, and who also prove the matching lower bound, so that $D_r(N) \sim \sqrt{rN}$ [CMT15, Thm. 3.10]. Schmutz and Tait [ST25, Thm. 3] later proved the same asymptotic formula independently.
- *The lower bound for sums* is the case $h = 2$ of a theorem of Cilleruelo, Ruzsa and Trujillo on $B_h[g]$ sets [CRT02], which (in the formulation of [Pla01, eq. (6)]) gives $S_r(N) \geq ((r+s)/\sqrt{r+2s} - o(1))\sqrt{N}$. It lifts a Singer difference set [Sin38] along a pattern of "layers".
- *The observation* that these two results settle Problem #863 was made by Boon Suan Ho with GPT-5.4 Pro in April 2026 [Ho26, Thr]; the problem page [Blo] now records it. The only additions here are the explicit error term in Proposition 3.1, the general form of the lifting lemma (Lemma 4.3), and the survey in Section 6.

1.4. **Why sums should beat differences.** A heuristic explains the direction of the inequality. Let $X \subseteq \{1,\dots,N\}$ be a random set with density δ. A positive difference d has about $\delta^2(N-d)$ representations $b - b'$, so the busiest differences are the short ones, with about $\delta^2 N$ representations each. A sum m has about $\delta^2 \min(m, 2N-m)$ ordered representations, so the busiest sums, those near N, also have about $\delta^2 N$ ordered representations, but only about $\delta^2 N/2$ *unordered* ones. Thus the constraint "at most r" allows density $\sqrt{r/N}$ for differences but $\sqrt{2r/N}$ for sums. The asymmetry is that $a + a'$ and $a' + a$ are the same sum, while $b - b'$ and $b' - b$ are different differences. For $r = 1$ this heuristic is wrong ($c_1 = c'_1$), since a single representation is far too few for averages to be meaningful; but as $r \to \infty$ it is essentially right: as $r \to \infty$ the ratio of $\liminf_N S_r(N)/\sqrt{N}$ to c'_r tends to a constant between 1.62 and 1.78 (see

Section 6). For fixed small r one needs explicit constructions, and the one used here gains a factor of only about 1.03–1.06.

1.5. **The idea of the proofs.** *Differences* (Section 3). Slide a window of length H, with $1 \ll H \ll N$, across $\{1, \dots, N\}$. Two elements of B in a common window differ by less than H, and there are only $H-1$ such differences, each available at most r times. Counting pairs of elements sharing a window from below (by Cauchy–Schwarz) and from above (by the multiplicity bound) shows that B cannot have average density more than about $\sqrt{r/N}$; see Fig. 1.

Sums (Section 4). Nothing like this holds for sums: two nearby elements can have a sum anywhere. The construction exploits this. Take a Singer set C modulo $Q = q^2+q+1$, a set of $q+1$ residues whose nonzero differences are all distinct, and place copies of it in several consecutive blocks ("layers") of length Q, using only the layers u in a fixed finite pattern P. Because C is a Sidon set modulo Q, a sum determines the residues of its two summands, and the representation count of the lifted set reduces to the number of ways of writing a layer index as $u+v$ with $u, v \in P$. A pattern $P \subseteq \{0, \dots, K-1\}$ with at most r such ordered representations yields $S_r(N) \gtrsim (|P|/\sqrt{K})\sqrt{N}$. An interval $P = \{0, \dots, r-1\}$ gives only $\sqrt{r}$; the pattern of Cilleruelo, Ruzsa and Trujillo adds s further layers after a gap of s empty layers, and gets $(r+s)/\sqrt{r+2s}$ (Figs. 2 and 3).

2. Notation and conventions

Write $[N] = \{1, \dots, N\}$. For a finite set $A \subseteq \mathbb{Z}$ and $m \in \mathbb{Z}$ let

$$R_A(m) = \#\{(a, a') \in A^2 : \ a \le a', \ a + a' = m\}$$

be the number of *unordered* representations of m as a sum of two elements of A (repetition $a = a'$ allowed), and for a finite set $B \subseteq \mathbb{Z}$ and an integer $d \ge 1$ let

$$\Delta_B(d) = \#\{(b, b') \in B^2 : \ b - b' = d\}$$

be the number of representations of d as a difference. Define

$$S_r(N) = \max\{|A| : \ A \subseteq [N], \ R_A(m) \le r \text{ for all } m \in \mathbb{Z}\},$$
$$D_r(N) = \max\{|B| : \ B \subseteq [N], \ \Delta_B(d) \le r \text{ for all } d \ge 1\}.$$

Both are nondecreasing in N, since the conditions are inherited by subsets and do not depend on the ambient interval. These are the conventions of the problem page [Blo]; some remarks on them:

(i) Sums are counted without order, as for Sidon sets. With ordered counting the representations $a + a'$ and $a' + a$ would always both be counted, and for $r = 1$ only one-element sets would qualify.
(ii) Only positive differences are restricted: $0 = b - b$ has $|B|$ representations. A positive difference $d = b - b'$ automatically has $b > b'$, so there is no ordered/unordered distinction to make.
(iii) Erdős takes $b_1 < \cdots < b_\ell < n$, which replaces $D_r(n)$ by $D_r(n-1)$; this does not affect any asymptotic statement.
(iv) Erdős writes the maxima as $(c_r + o(1))\sqrt{n}$ and $(c'_r + o(1))\sqrt{n}$, presupposing that the limits exist. Theorem 1.1 does not need this.

3. Differences: the Erdős–Turán argument

The following is the Erdős–Turán argument [ET41], in a form with overlapping windows; for $r = 1$ it is their bound $\sqrt{N} + O(N^{1/4})$ for Sidon sets. (See O'Bryant's survey [OB04] for the many later refinements of this argument for Sidon sets, and [CMT15, ST25] for two other proofs of the same bound for general r.)

Proposition 3.1. *For all integers $r, N \geq 1$,*

$$D_r(N) < \sqrt{rN} + \Big(\frac{\sqrt{r}}{2} + 2\Big)N^{1/4}.$$

In particular $\limsup_{N\to\infty} D_r(N)/\sqrt{N} \leq \sqrt{r}$.

Proof. Let $B \subseteq [N]$ satisfy $\Delta_B(d) \leq r$ for all $d \geq 1$, and put $M = |B|$.

A crude bound. The $\binom{M}{2}$ pairs $b > b'$ in B have differences in $\{1, \dots, N-1\}$, each value occurring at most r times, so $M(M-1) \leq 2r(N-1) < 2rN$. This forces $M \leq 2\sqrt{rN}$: otherwise $M(M-1) > 2\sqrt{rN}\,(2\sqrt{rN} - 1) = 4rN - 2\sqrt{rN} \geq 2rN$, since $rN \geq 1$.

Windows. Fix an integer H with $1 \leq H \leq N$. For each integer x in the range $1 - H \leq x \leq N - 1$ (there are $N + H - 1$ of them) let

$$I_x = \{x + 1, x + 2, \dots, x + H\}, \qquad w_x = |B \cap I_x|.$$

An element $b \in [N]$ lies in I_x exactly when $b - H \leq x \leq b - 1$, that is, in exactly H of the windows. Hence $\sum_x w_x = MH$, and by the Cauchy–Schwarz inequality

$$\sum_x w_x(w_x - 1) = \sum_x w_x^2 - \sum_x w_x \;\geq\; \frac{M^2H^2}{N + H - 1} - MH. \tag{1}$$

Counting pairs in a window. The left side of (1) counts triples (x, b, b') with $b \neq b'$ both in I_x. If $b > b'$ and $d = b - b'$, then both lie in I_x exactly when $b - H \leq x \leq b' - 1$; this happens for $H - d$ values of x if $d < H$, and for none if $d \geq H$ (see Fig. 1). Counting both orders of each pair,

$$\sum_x w_x(w_x - 1) = 2\sum_{d=1}^{H-1} (H - d)\,\Delta_B(d) \;\leq\; 2r\sum_{d=1}^{H-1}(H - d) = rH(H - 1). \tag{2}$$

Conclusion. Combining (1) and (2) and multiplying by $(N + H - 1)/H^2$ gives

$$M^2 \;\leq\; r\Big(1 - \frac{1}{H}\Big)(N + H - 1) + M\,\frac{N + H - 1}{H}. \tag{3}$$

Now take $H = \lceil N^{3/4} \rceil$, so that $N^{3/4} \leq H < N^{3/4} + 1$ and $H \leq N$. The first term of (3) is less than $r(N + N^{3/4})$. In the second, $(N + H - 1)/H < 2N/H \leq 2N^{1/4}$, and $M \leq 2\sqrt{rN}$, so the second term is at most $4\sqrt{r}\,N^{3/4}$. Therefore

$$M^2 < rN + (r + 4\sqrt{r})N^{3/4},$$

and since $\sqrt{a + b} \leq \sqrt{a} + b/(2\sqrt{a})$ for $a > 0$, $b \geq 0$,

$$M < \sqrt{rN} + \frac{(r + 4\sqrt{r})N^{3/4}}{2\sqrt{rN}} = \sqrt{rN} + \Big(\frac{\sqrt{r}}{2} + 2\Big)N^{1/4}. \qquad \square$$

Remark 3.2. Only differences $d < H$, that is $d \leq N^{3/4}$, enter (2); the long differences are used only in the crude bound $M \leq 2\sqrt{rN}$. This is the precise sense in which the difference problem is *local*: a set with few repeated short differences cannot be denser than $\sqrt{r/N}$ on average.

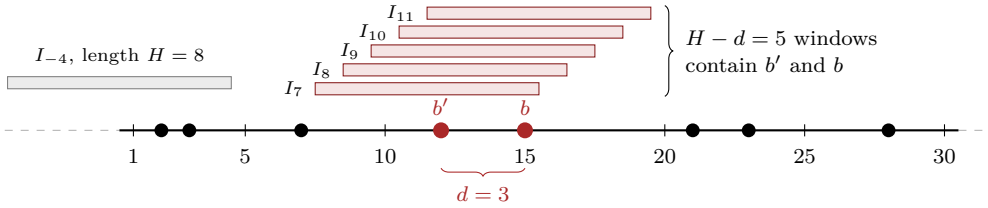


Figure 1. The window count in the proof of Proposition 3.1, with $N = 30$, $H = 8$ and $B = \{2, 3, 7, 12, 15, 21, 23, 28\}$. The windows $I_x = \{x + 1, \dots, x + H\}$, $1 - H \le x \le N - 1$, may stick out of $[N]$, so that every element of B lies in exactly H of them. Two elements at distance $d < H$ lie together in exactly $H - d$ windows (here $I_7, \dots, I_{11}$). Since every short difference d occurs at most r times, few pairs can share a window, while Cauchy–Schwarz forces many pairs to share one if B is large.

Better error terms are known: Lindström [Lin69] proved $\sqrt{N} + N^{1/4} + 1$ for Sidon sets, and [CMT15, Thm. 2.5] gives $\sqrt{rN} + (rN)^{1/4} + 1$.

4. Sums: lifting a Singer set

4.1. Cyclic Sidon sets.

The building block is Singer's construction.

Theorem 4.1 (Singer [Sin38])**.** *Let q be a prime power and $Q = q^2 + q + 1$. There is a set $C \subseteq \mathbb{Z}/Q\mathbb{Z}$ with $|C| = q + 1$ such that every nonzero residue modulo Q has exactly one representation $c - c'$ with $c, c' \in C$.*

Such a set is a *planar* (or perfect) difference set: its translates are the lines of the projective plane $\mathrm{PG}(2, q)$, cyclically permuted by a Singer cycle. For example, $\{0, 1, 3\}$ modulo 7 and $\{0, 1, 3, 9\}$ modulo 13. Bose [Bos42] gave a similar construction of q residues modulo $q^2 - 1$ with distinct differences, which would serve equally well below. The only property we need is that the nonzero differences are distinct, which makes C a Sidon set for sums modulo Q:

Lemma 4.2. *Let $C \subseteq \mathbb{Z}/Q\mathbb{Z}$ be a set whose nonzero differences $c - c'$ $(c \ne c')$ are pairwise distinct. If $c_1 + c_2 = c_3 + c_4$ in $\mathbb{Z}/Q\mathbb{Z}$ with all $c_i \in C$, then $\{c_1, c_2\} = \{c_3, c_4\}$ as multisets.*

Proof. If $c_1 = c_3$ then $c_2 = c_4$. Otherwise $c_1 - c_3 = c_4 - c_2$ is a nonzero difference represented by the pairs (c_1, c_3) and (c_4, c_2), so these pairs coincide: $c_1 = c_4$ and $c_3 = c_2$. □

4.2. Lifting along a layer pattern.

For a finite set P of nonnegative integers (the *layer pattern*) and $t \in \mathbb{Z}$, write

$$\rho_P(t) = \#\{(u, v) \in P^2 : \ u + v = t\}$$

for the number of *ordered* representations of t as a sum of two elements of P.

Lemma 4.3 (Lifting)**.** *Let $P \subseteq \{0, 1, \dots, K - 1\}$ satisfy $\rho_P(t) \le r$ for all t, and let $C \subseteq \{0, 1, \dots, Q - 1\}$ be a set of distinct residues modulo Q whose nonzero differences are distinct modulo Q. Then the set*

$$A = \{uQ + c + 1 : \ u \in P, \ c \in C\}$$

satisfies $A \subseteq [KQ]$, $|A| = |P|\,|C|$, and $R_A(m) \le r$ for every integer m.

Proof. Every $x \in A$ has a unique expression $x = uQ + c + 1$ with $u \in P$, $c \in C$: u and c are the quotient and remainder of $x - 1$ on division by Q. We call u the *layer* and c the *residue* of x. This gives $|A| = |P||C|$, and $1 \le x \le (K - 1)Q + (Q - 1) + 1 = KQ$.

Fix m with $R_A(m) > 0$, and consider any representation $m = x + y$ with $x = uQ + c + 1$, $y = vQ + d + 1$ in A. Reducing modulo Q, $c + d \equiv m - 2$. By Lemma 4.2 the multiset $\{c, d\}$

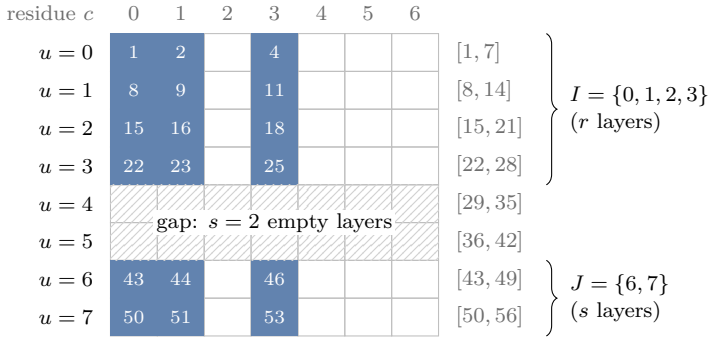


Figure 2. The lifted set of Lemma 4.3 for $r=4$ (so $s=2$, $P=P_4=\{0,1,2,3,6,7\}$, $K=8$) and the Singer set $C=\{0,1,3\}$ modulo $Q=7$ ($q=2$). Row u is the block $\{uQ+1,\dots,uQ+Q\}$ of $[KQ]=[56]$; the shaded cells are the 18 elements of $A=\{uQ+c+1\}$. Because C is a Sidon set modulo 7, the residues of two summands are determined by their sum, and the number of representations of m is at most the number of ordered layer pairs $(u,v)\in P^2$ with $u+v=T$. A direct computation confirms that every integer has at most 4 representations $a+a'$, $a\le a'$, in this A.

is therefore the same, say $\{c_0,d_0\}$, for every representation of m; consequently the integer $c+d=c_0+d_0$ is the same, and so is the layer sum

$$u+v=T:=\frac{m-2-c_0-d_0}{Q}.$$

Case $c_0\neq d_0$. In every representation $\{x,y\}$ exactly one summand has residue c_0. Map the representation to the pair (layer of the summand with residue c_0, layer of the summand with residue d_0). This is a pair $(u,v)\in P^2$ with $u+v=T$, and it determines both summands, so the map is injective. Hence $R_A(m)\le\rho_P(T)\le r$.

Case $c_0=d_0$. Now the representations are exactly the pairs $\{uQ+c_0+1,\ vQ+c_0+1\}$ with $u,v\in P$, $u\le v$, $u+v=T$, so $R_A(m)=\#\{(u,v)\in P^2:u\le v,\ u+v=T\}\le\rho_P(T)\le r$. □

Corollary 4.4. *If* $P\subseteq\{0,1,\dots,K-1\}$ *satisfies* $\rho_P(t)\le r$ *for all* t, *then*

$$\liminf_{N\to\infty}\frac{S_r(N)}{\sqrt{N}}\ \ge\ \frac{|P|}{\sqrt{K}}.$$

Proof. Let $\varepsilon>0$ and put $X=\sqrt{N/K}$. By the prime number theorem, for all large N there is a prime q with $(1-\varepsilon)(X-1)<q\le X-1$. Then $Q=q^2+q+1\le(X-1)^2+(X-1)+1=X^2-X+1\le X^2$, so $KQ\le N$. Applying Lemma 4.3 with a Singer set C (Theorem 4.1) and using monotonicity,

$$S_r(N)\ge S_r(KQ)\ge|P|(q+1)>|P|((1-\varepsilon)(X-1)+1)\ge(1-\varepsilon)\frac{|P|}{\sqrt{K}}\sqrt{N}.$$

As ε was arbitrary, the claim follows. □

4.3. **The pattern of Cilleruelo, Ruzsa and Trujillo.** It remains to choose a good pattern. An interval $P=\{0,\dots,r-1\}$ has $\rho_P\le r$ and gives only $|P|/\sqrt{K}=\sqrt{r}$, the same constant as for differences. The point is that the sumset $P+P$ of an interval is a "tent" that reaches height r only in the middle; there is room left in its tails. Adding a second block of s layers after a gap of s layers fills part of the room in the right tail without exceeding height r (Fig. 3).

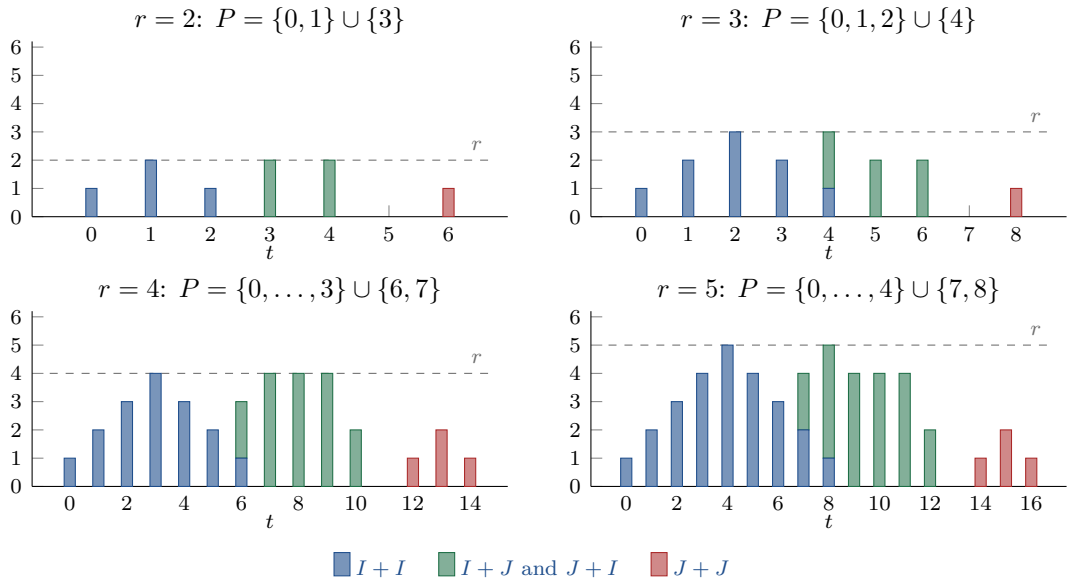


Figure 3. The ordered representation counts $\rho_{P_r}(t)$ of the pattern of Lemma 4.5 for $r = 2, 3, 4, 5$, split into the contributions of $I + I$ (blue), of $I + J$ and $J + I$ together (green) and of $J + J$ (red). The dashed line is height r. The block J is placed so that the cross sums begin exactly where the tent $I + I$ has descended far enough to leave room for them, and $J + J$ lies beyond everything else. In each case the maximum is exactly r, so the pattern is tight.

Lemma 4.5 ([CRT02])**.** *Let $r \geq 1$, $s = \lfloor r/2 \rfloor$, and*

$$P_r = I \cup J, \qquad I = \{0, 1, \ldots, r-1\}, \quad J = \{r+s,\, r+s+1, \ldots, r+2s-1\}.$$

Then $|P_r| = r + s$, $P_r \subseteq \{0, 1, \ldots, r + 2s - 1\}$, and $\rho_{P_r}(t) \leq r$ for all t.

Proof. The first two claims are clear (I and J are disjoint as $r \leq r + s$). For finite sets X, Y write $\rho_{X,Y}(t) = \#\{(x, y) \in X \times Y : x + y = t\}$; then

$$\rho_{P_r} = \rho_{I,I} + 2\rho_{I,J} + \rho_{J,J}.$$

The three terms are supported on $[0, 2r-2]$, $[r+s, 2r+2s-2]$ and $[2r+2s, 2r+4s-2]$ respectively, so the support of $\rho_{J,J}$ is disjoint from the other two. We have $\rho_{I,I}(t) \leq r$, $\rho_{J,J}(t) \leq |J| = s \leq r$ and $2\rho_{I,J}(t) \leq 2|J| = 2s \leq r$, which proves the claim for every t outside the overlap $[r+s, 2r-2]$ of the first two supports.

This overlap is nonempty only if $s \leq r - 2$; write its elements as $t = r + s + u$ with $0 \leq u \leq r-s-2$. Since $t \geq r-1$, the tent $\rho_{I,I}$ is on its descending side: $\rho_{I,I}(t) = 2r-1-t = r-s-1-u$. Next, $\rho_{I,J}(t)$ is the number of pairs (x, j) with $x + j = u$, $0 \leq x \leq r - 1$, $0 \leq j \leq s - 1$ (put $y = r + s + j \in J$); since $u \leq r - 1$, this is the number of $j \in [0, s - 1]$ with $j \leq u$, namely $\min(u + 1, s)$. Hence, using $2\min(a, b) \leq a + b$,

$$\rho_{P_r}(t) = (r - s - 1 - u) + 2\min(u + 1, s) \leq (r - s - 1 - u) + (u + 1) + s = r. \qquad \square$$

Proposition 4.6 (Cilleruelo–Ruzsa–Trujillo [CRT02])**.** *For every $r \geq 1$, with $s = \lfloor r/2 \rfloor$,*

$$\liminf_{N\to\infty} \frac{S_r(N)}{\sqrt{N}} \;\geq\; \frac{r+s}{\sqrt{r+2s}}.$$

Proof. Apply Corollary 4.4 to the pattern P_r of Lemma 4.5, with $K = r + 2s$. $\square$

Remark 4.7. Corollary 4.4 applies to any pattern. For $r = 2$ the condition $\rho_P \le 2$ says exactly that P is a Sidon set, and the Sidon set $\{0,1,4,6\}$ gives $4/\sqrt{7} \approx 1.5118$, better than the value $3/2$ from $P_2 = \{0,1,3\}$. This is the bound of Habsieger and Plagne [HP02], who also showed that no pattern does better for $r = 2$ [Pla01, §2.2]. Plagne [Pla01] found patterns improving on Proposition 4.6 for some other small r: for $r = 4$ the 12-element pattern $\{0,1,3,9,10,11,13,18,24,25,29,30\}$ has $\rho_P \le 4$ and gives $12/\sqrt{31} \approx 2.1552$, and for $r = 6$ the pattern $\{0,1,2,3,4,5,8,9,12,14,18,19\}$ gives $12/\sqrt{20} \approx 2.6832$ (we have checked both conditions $\rho_P \le r$ by computer).

5. Proof of the main theorem

Proof of Theorem 1.1. The first inequality is Proposition 3.1 and the last is Proposition 4.6. For the middle one,

$$\Big(\frac{r+s}{\sqrt{r+2s}}\Big)^2 - r = \frac{(r+s)^2 - r(r+2s)}{r+2s} = \frac{s^2}{r+2s} > 0,$$

because $s = \lfloor r/2 \rfloor \ge 1$ when $r \ge 2$. □

Proof of Corollary 1.2. If the limits exist, then by Theorem 1.1

$$c'_r = \limsup_{N\to\infty} \frac{D_r(N)}{\sqrt{N}} \le \sqrt{r} < \liminf_{N\to\infty} \frac{S_r(N)}{\sqrt{N}} = c_r.$$ □

For $r = 1$ the argument gives nothing, as it must: $s = 0$, $P_1 = \{0\}$, and both bounds equal 1.

6. What is known about the two constants

6.1. **Differences.** The constant c'_r is known exactly.

Theorem 6.1 ([CMT15, Thm. 3.10]; [ST25, Thm. 3])**.** *For every fixed $r \ge 1$, $D_r(N) = (1+o(1))\sqrt{rN}$ as $N \to \infty$. Thus c'_r exists and equals $\sqrt{r}$.*

The upper bound is Proposition 3.1. The lower bound comes from projecting a Bose set; we sketch the argument of [CMT15, ST25] for completeness. Let q be a prime with $q \equiv 1 \pmod r$. Bose [Bos42] constructed a set $\mathcal{B} \subseteq \mathbb{Z}/(q^2-1)\mathbb{Z}$ of q residues whose $q(q-1)$ nonzero differences are distinct and are precisely the residues not divisible by $q+1$. Put $M = (q^2-1)/r$; since $r \mid q-1$, M is a multiple of $q+1$, so the kernel of the reduction map $\pi : \mathbb{Z}/(q^2-1)\mathbb{Z} \to \mathbb{Z}/M\mathbb{Z}$ consists of multiples of $q+1$. Hence π is injective on $\mathcal{B}$ (no nonzero difference of $\mathcal{B}$ lies in the kernel), and for each nonzero $\delta \in \mathbb{Z}/M\mathbb{Z}$ the r residues in $\pi^{-1}(\delta)$ are each a difference of $\mathcal{B}$ at most once, so δ has at most r representations as a difference in $\pi(\mathcal{B})$. Taking representatives in $[M]$, every $d \in [1, M-1]$ is nonzero modulo M, so $D_r(M) \ge q \approx \sqrt{rM}$; the prime number theorem for the progression $1 \bmod r$ fills in the remaining N. (We checked this construction numerically for primes $q \le 13$.) Xu [Xu18] had earlier obtained results on popular differences from which the upper bound, and the limiting behaviour as $r \to \infty$, can be deduced [ST25].

6.2. **Sums.** In the language of [CRV10], a set $A \subseteq \mathbb{Z}$ with $R_A \le r$ is the same as a set whose *ordered* representation function is at most $2r$ (in $\mathbb{Z}$ the ordered count of m is $2R_A(m)$ or $2R_A(m)-1$), so $S_r(N)$ is the quantity $\beta_{2r}(N)$ of [CRV10], and the quantity $F_{2,r}(N)$ of [Pla01, HP18]. It is not known whether $S_r(N)/\sqrt{N}$ converges for any $r \ge 2$ [CRV10, Pla01]. The known bounds include:

- *Trivial upper bound.* The $\binom{|A|+1}{2}$ unordered pairs have sums in $[2, 2N]$, so $\binom{|A|+1}{2} \le r(2N-1)$ and $\limsup S_r(N)/\sqrt{N} \le 2\sqrt{r}$.

- *Better upper bounds* come from Fourier analysis; Habsieger and Plagne [HP18] prove $\limsup S_r(N)/\sqrt{N} \le \sqrt{1.740463\,(2r-1)}$, which improved the previous records for $2 \le r \le 5$; for $r = 2$ it gives 2.2851.
- *Lower bounds* for small r come from glued constructions as in Lemma 4.3: Proposition 4.6 and the improvements in Remark 4.7. Habsieger and Plagne conjectured that $c_2 = 2$ [Pla01].
- *Large r.* Cilleruelo, Ruzsa and Vinuesa [CRV10] proved that

$$\lim_{r\to\infty} \liminf_{N\to\infty} \frac{S_r(N)}{\sqrt{2rN}} = \lim_{r\to\infty} \limsup_{N\to\infty} \frac{S_r(N)}{\sqrt{2rN}} = \sigma,$$

  where σ is the supremum of $\int_0^1 f$ over nonnegative f on $[0,1]$ whose autoconvolution $f*f$ is at most 1 everywhere; the bounds $1.1509 \le \sigma \le 1.2525$ are due to Matolcsi and Vinuesa [MV10]. Since $c'_r = \sqrt{r}$, the ratio of the two constants for large r is about $\sqrt{2}\,\sigma \in [1.627, 1.772]$, much more than the factor ≈ 1.06 given by Theorem 1.1.

Table 1. Constants for small r. The second column is c'_r (Theorem 6.1). The third is the lower bound of Proposition 4.6, and the fourth its ratio to c'_r. The last column is the upper bound $\sqrt{1.740463(2r-1)}$ of [HP18]. Lower bounds are rounded down and upper bounds up. For $r = 2$ the lower bound improves to $4/\sqrt{7} \approx 1.5118$ [HP02]; see Remark 4.7 for others.

r	$c'_r = \sqrt{r}$	$\frac{r+s}{\sqrt{r+2s}} \le \liminf \frac{S_r(N)}{\sqrt{N}}$	ratio	$\limsup \frac{S_r(N)}{\sqrt{N}} \le$
2	1.4142	1.5000	1.0606	2.2851
3	1.7321	1.7888	1.0327	2.9500
4	2.0000	2.1213	1.0606	3.4905
5	2.2361	2.3333	1.0434	3.9578

Acknowledgements and provenance

The observation that Problem #863 follows from the literature, and the proofs in the original note [Ho26], were found by GPT-5.4 Pro prompted by Boon Suan Ho, who announced them on the erdosproblems.com forum on 22 April 2026 [Thr]; Nat Sothanaphan reported on the same thread that a check of the note found no issues, and the problem page [Blo] was updated accordingly. The papers [CMT15, ST25] on r-Golomb rulers, which already contain $c'_r = \sqrt{r}$, are not mentioned in the original note or on the problem page. This text was written by Claude Opus 5.5.

References

[Blo] T. F. Bloom, *Erdős Problem #863*, https://www.erdosproblems.com/863, accessed 28 October 2026.
[Bos42] R. C. Bose, *An affine analogue of Singer's theorem*, J. Indian Math. Soc. (N.S.) **6** (1942), 1–15.
[CMT15] Y. Caicedo, C. A. Martos, and C. A. Trujillo, *g-Golomb rulers*, Rev. Integr. Temas Mat. **33** (2015), no. 2, 161–172, doi:10.18273/revint.v33n2-2015006.
[CRT02] J. Cilleruelo, I. Z. Ruzsa, and C. Trujillo, *Upper and lower bounds for finite $B_h[g]$ sequences*, J. Number Theory **97** (2002), no. 1, 26–34, doi:10.1006/jnth.2001.2767.
[CRV10] J. Cilleruelo, I. Z. Ruzsa, and C. Vinuesa, *Generalized Sidon sets*, Adv. Math. **225** (2010), no. 5, 2786–2807.
[Er92] P. Erdős, *Some of my forgotten problems in number theory*, Hardy–Ramanujan J. **15** (1992), 34–50, doi:10.46298/hrj.1992.125.
[EF91] P. Erdős and R. Freud, *A Sidon-problémakör* [On Sidon sequences and related problems], Mat. Lapok 1991/2, 1–44 (in Hungarian).

[ET41] P. Erdős and P. Turán, *On a problem of Sidon in additive number theory, and on some related problems*, J. London Math. Soc. **16** (1941), 212–215.
[HP02] L. Habsieger and A. Plagne, *Ensembles $B_2[2]$: l'étau se resserre*, Integers **2** (2002), Paper A2.
[HP18] L. Habsieger and A. Plagne, *A numerical note on upper bounds for $B_2[g]$ sets*, Exp. Math. **27** (2018), no. 2, 208–214; arXiv:1609.02771.
[Ho26] B. S. Ho, *On a problem of Erdős, Berend, and Freud concerning bounded sums and bounded differences*, note, April 2026; see the version note above.
[Lin69] B. Lindström, *An inequality for B_2-sequences*, J. Combin. Theory **6** (1969), 211–212.
[MV10] M. Matolcsi and C. Vinuesa, *Improved bounds on the supremum of autoconvolutions*, J. Math. Anal. Appl. **372** (2010), no. 2, 439–447.
[OB04] K. O'Bryant, *A complete annotated bibliography of work related to Sidon sequences*, Electron. J. Combin. (2004), Dynamic Survey DS11.
[Pla01] A. Plagne, *Recent progress on finite $B_h[g]$ sets*, Congr. Numer. **153** (2001), 49–64.
[ST25] E. Schmutz and M. Tait, *Cardinalities of g-difference sets*, Integers **25** (2025), Paper A77.
[Sin38] J. Singer, *A theorem in finite projective geometry and some applications to number theory*, Trans. Amer. Math. Soc. **43** (1938), no. 3, 377–385.
[Thr] Erdős Problems forum, *Discussion thread for Problem #863*, https://www.erdosproblems.com/forum/thread/863, accessed 28 October 2026.
[Xu18] M. W. Xu, *Popular differences and generalized Sidon sets*, J. Number Theory **186** (2018), 103–120.

Anthropic